

MATHEMATICAL INDUCTION

- **Mathematical Induction Principle:** Let $P(n)$ be any statement where $n \in N$,
 - (a) Prove that $P(1)$ is true.
 - (b) Let us assume that $P(k)$ be true, $k \leq n \in N$.
 - (c) Prove that $P(k+1)$ is true whenever $P(k)$ is true.
 - (d) If $P(k+1)$ is true then $P(n)$ is true for all values of n .
- **Fermat Little Theorem:** $n^p - n$ is divisible by p where n is any integer and p is a prime.
- If n is not divisible by p , then $n^{p-1} - 1$ is divisible by p where n is any integer and p is a prime.
- For any prime p , $(p-1)!+1$ is divisible by p .
